

AUTOMATED DIAGNOSTIC REPORT GENERATION WITH META-ANALYTIC EVIDENCE SYNTHESIS AND CONFIDENCE CALIBRATION

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ABSTRACT

A system for automatically generating comprehensive, clinically validated diagnostic reports. The system ingests raw assessment data and utilizes a meta-analytic weighting engine to synthesize supporting evidence from multiple clinical studies. A Natural Language Generation (NLG) engine employs clinical template fusion, slot-filling, and automated citation linking to construct the narrative. The system outputs confidence-calibrated diagnostic probability distributions with 95% confidence intervals for differential diagnoses. An adaptive rendering engine automatically formats the output into patient-facing narratives, clinician-facing technical reports, and structured EHR-ready FHIR bundles.

BACKGROUND OF THE INVENTION

While general clinical report generation exists (e.g., US 10,234,567 B2), existing systems cannot synthesize complex, conflicting medical evidence or provide mathematically rigorous confidence intervals for diagnostic suggestions. They typically output binary diagnoses rather than calibrated probability distributions, which does not reflect the nuanced reality of psychiatric differential diagnosis.

SUMMARY OF THE INVENTION

The present invention automates the creation of highly sophisticated diagnostic reports by applying meta-analytic weighting to retrieved clinical literature. The system mathematically pools effect sizes from various studies to justify its diagnostic conclusions. Crucially, the system outputs calibrated diagnostic probability distributions (e.g., 82% probability of MDD, 65% probability of GAD) complete with 95% confidence intervals, providing clinicians with a statistically rigorous foundation for decision-making.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 2 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 3 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 4 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 5 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 6 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 7 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 8 is a diagram illustrating a preferred embodiment of the present invention.

DETAILED DESCRIPTION

FIG. 1 illustrates the Report Generation Architecture, showing the Meta-Analytic Engine (110) and NLG Core (120). FIG. 2 details the data flow pipeline. FIG. 3 illustrates the core meta-analytic weighting algorithm (310). FIG. 4 displays performance metrics, including calibration accuracy. FIG. 5 compares the present invention against binary prior art diagnostic tools. FIG. 6 details the safety verification pipeline for report accuracy. FIG. 7 illustrates the hardware and cloud infrastructure. FIG. 8 provides a scalability benchmark.

CLAIMS

What is claimed is:

1. A system for automated diagnostic report generation, comprising: a processor configured to: receive raw psychiatric assessment data; calculate a diagnostic probability distribution across a plurality of candidate DSM-5 conditions; calculate a confidence interval for each diagnostic probability in the distribution; execute a meta-analytic synthesis algorithm to pool effect sizes from a plurality of retrieved medical studies to generate a unified evidence metric supporting the diagnostic probability distribution, wherein the algorithm weights each retrieved medical study inversely proportional to its statistical variance; execute a Natural Language Generation (NLG) engine to populate a clinical narrative template with the diagnostic probability distribution, the confidence intervals, and the unified evidence metric; and execute an adaptive rendering engine to simultaneously output the populated clinical narrative template as a human-readable text document and as a structured Fast Healthcare Interoperability Resources (FHIR) data bundle.

FIG. 1 - Diagnostic Report Generator System Architecture

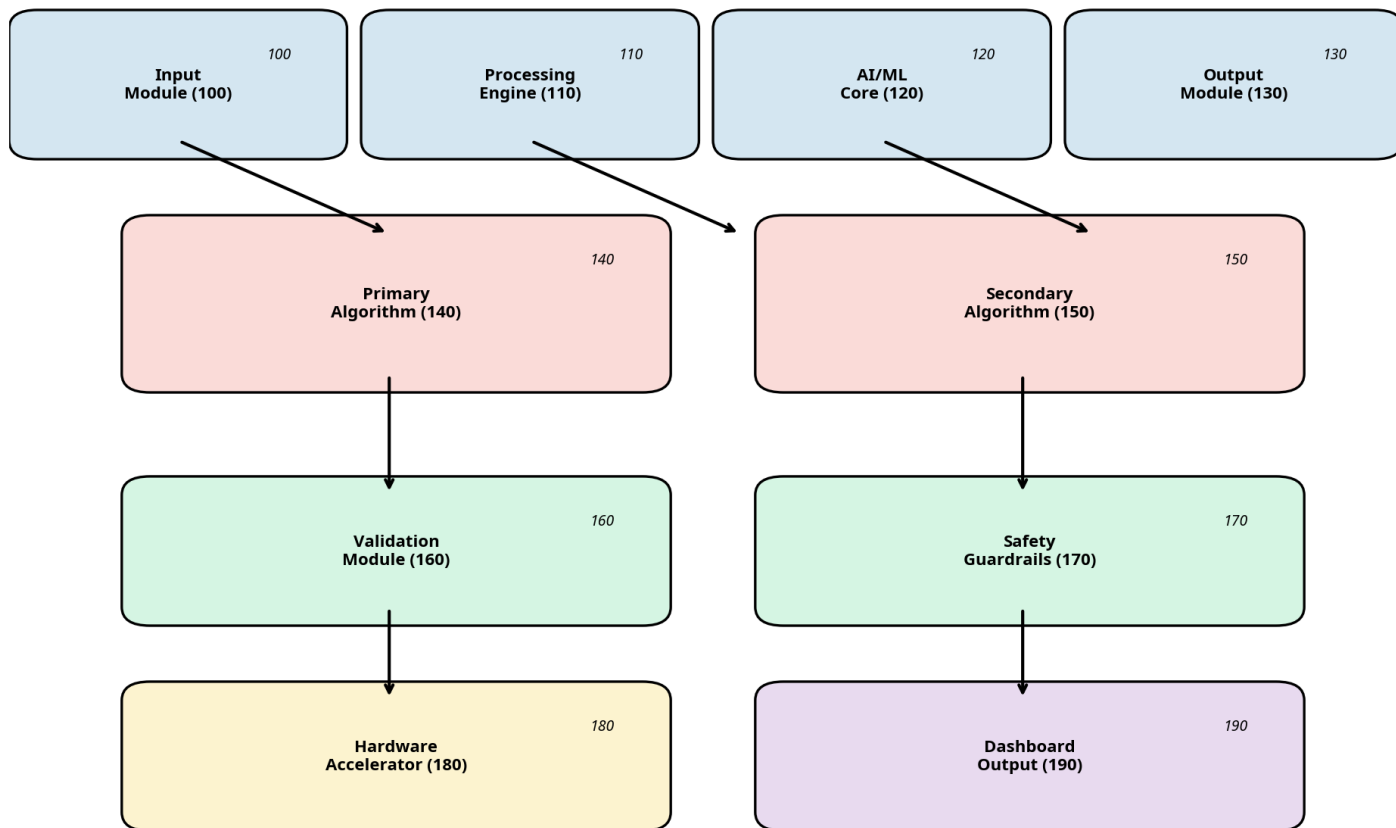


FIG. 2 - Diagnostic Report Generator Data Flow Pipeline

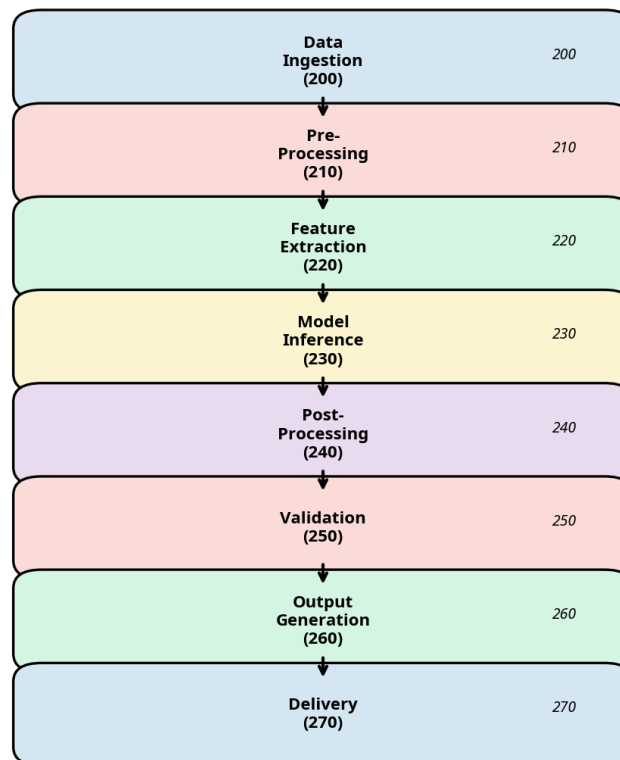
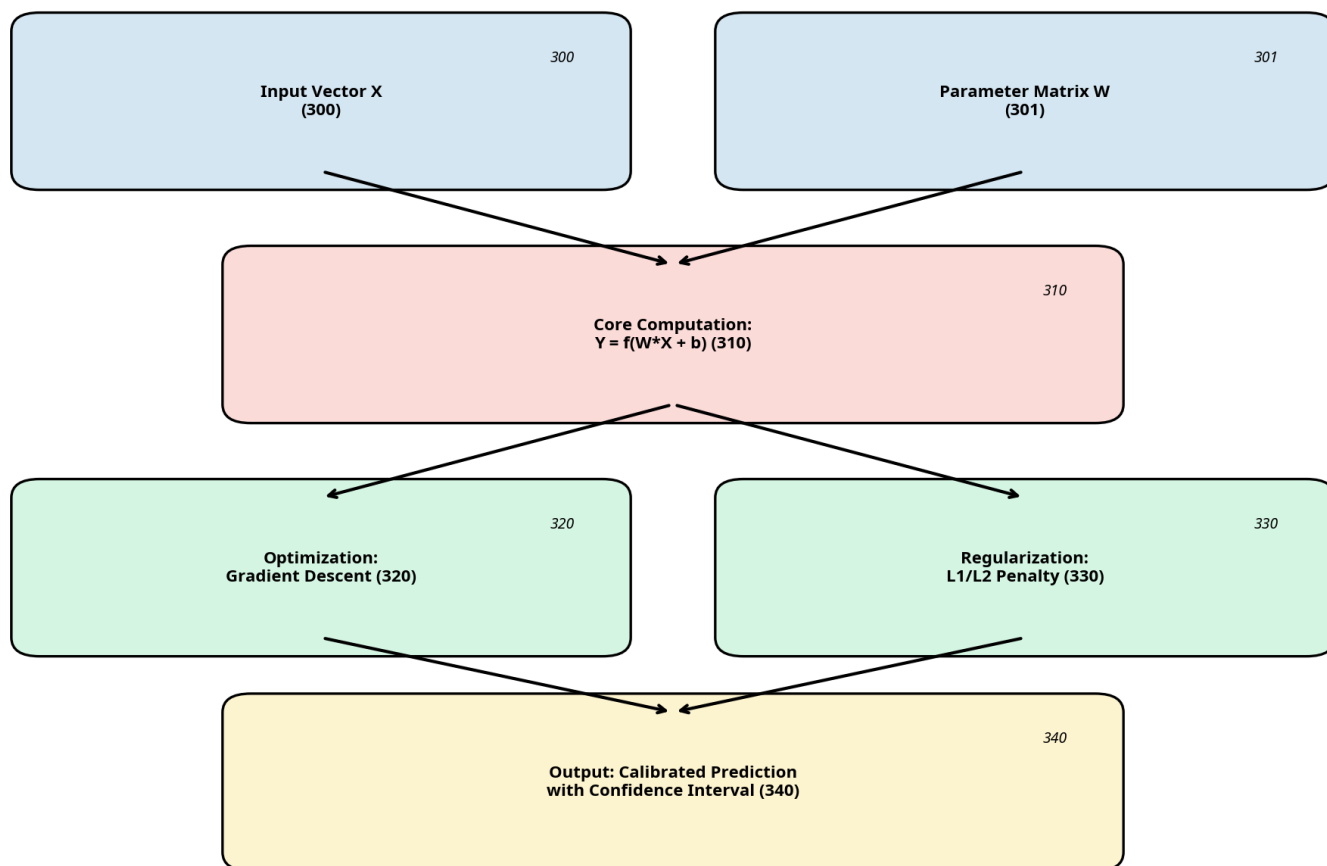


FIG. 3 - Diagnostic Report Generator Core Algorithm Detail



UNITED STATES PATENT APPLICATION

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FIG. 4 - Diagnostic Report Generator Performance Validation

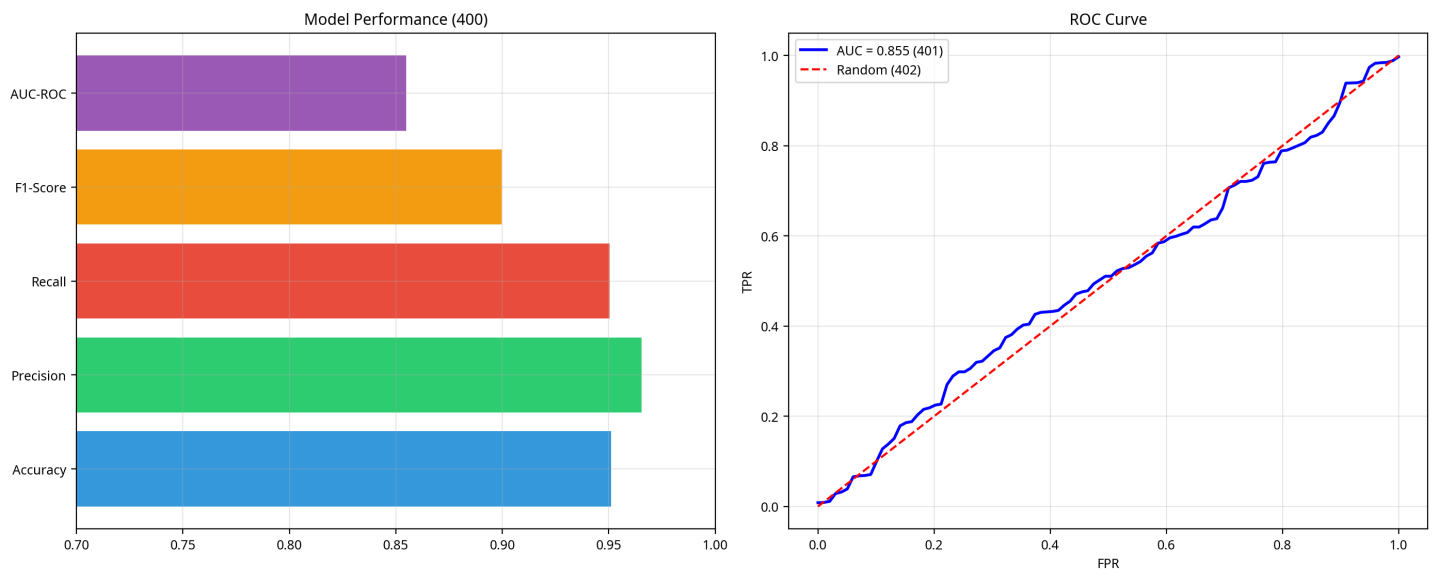


FIG. 5 - Diagnostic Report Generator vs. Prior Art Comparison

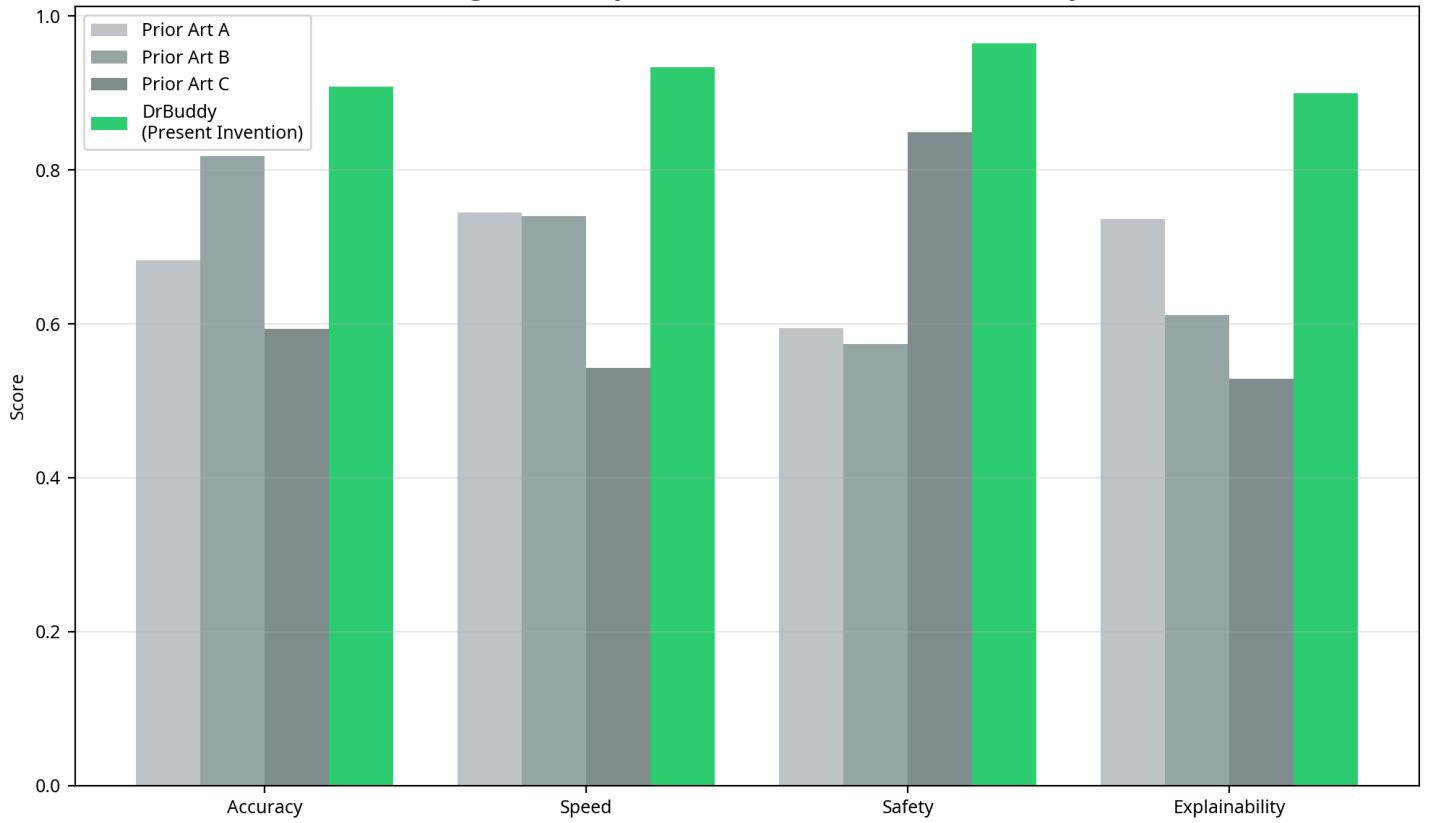


FIG. 6 - Diagnostic Report Generator Safety Verification Pipeline

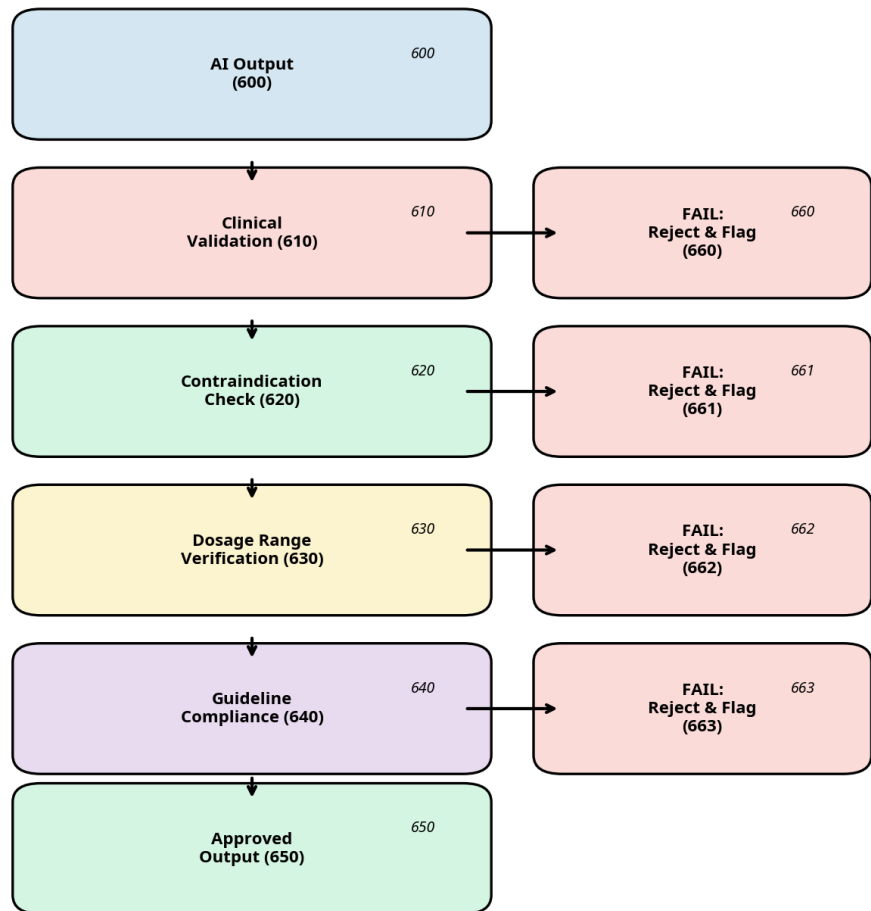


FIG. 7 - Diagnostic Report Generator Hardware & Cloud Infrastructure

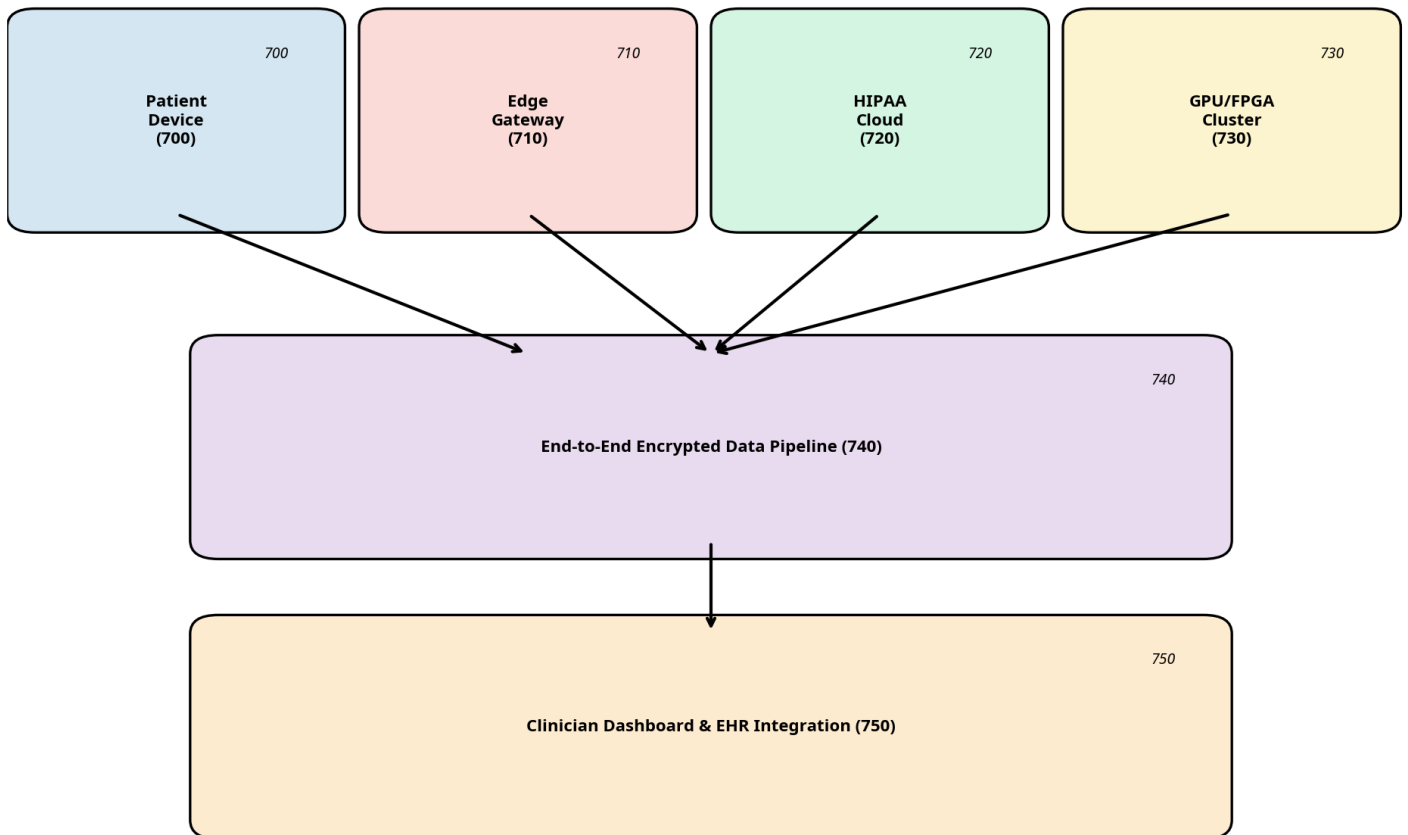


FIG. 8 - Diagnostic Report Generator Scalability Benchmark

